

BOULDER, CO, USA

WMO: 720533

Lat: 40.033N

Lon: 105.217W

Elev: 1612

StdP: 83.40

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 00160

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-15.2	-12.4	-21.2	0.7	-2.4	-18.8	0.9	-0.8	14.4	10.9	11.8	8.6	0.9	50	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	34.9	15.2	32.7	15.0	32.0	15.0	18.1	25.8	17.4	25.8	16.7	25.4	3.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	14.1	19.4	15.2	13.1	19.3	14.0	12.2	19.6	58.0	25.2	55.7	25.5	53.4	25.4	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	8.3	6.7	-19.8	37.3	3.3	1.1	-22.1	38.1	-24.0	38.8	-25.9	39.4	-28.3	40.2		
(5)			-19.6	19.5	3.4	1.0	-22.1	20.2	-24.1	20.7	-26.0	21.3	-28.5	22.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	1.7	1.4	7.0	9.8	13.3	21.1	23.7	22.9	19.0	11.1	5.7	1.0	(6)	(7)
	DBStd	9.46	5.43	6.46	5.59	4.67	5.04	3.66	2.41	2.70	4.04	5.38	5.39	5.74	(7)	(8)
	HDD10.0	1196	259	246	123	59	28	0	0	0	2	51	144	282	(8)	(9)
	HDD18.3	2968	516	473	353	257	166	14	1	4	40	227	379	538	(9)	(10)
	CDD10.0	1749	2	6	29	53	131	332	423	399	271	84	16	3	(10)	(11)
	CDD18.3	478	0	0	1	1	9	97	166	144	59	2	0	0	(11)	(12)
	CDH23.3	5452	0	0	12	22	158	1175	1728	1519	752	83	2	0	(12)	(13)
	CDH26.7	2253	0	0	1	2	33	506	766	653	284	9	0	0	(13)	(14)
(14) Wind	WSAvg	2.1	2.3	2.5	2.5	2.6	2.0	1.9	1.8	1.8	1.6	1.9	2.1	1.9	(14)	(15)
(15) Precipitation	PrecAvg	489	17	19	44	62	75	57	48	41	45	34	31	18	(15)	(16)
	PrecMax	748	56	62	138	153	244	153	132	141	124	137	88	55	(16)	(17)
	PrecMin	283	0	0	8	4	0	8	2	1	0	1	0	0	(17)	(18)
	PrecStd	122	12	13	31	39	53	35	29	33	32	29	21	13	(18)	(19)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	20.2	25.1	26.0	29.9	36.4	36.3	35.9	34.8	27.8	22.4	17.4	(19)	(20)
		MCWB	6.2	7.5	9.3	9.9	12.3	14.9	15.8	15.0	14.9	12.2	8.6	5.3	(20)	(21)
	2%	DB	15.1	17.1	22.1	22.9	27.3	33.8	34.8	33.0	32.2	26.1	19.8	14.9	(21)	(22)
		MCWB	4.6	5.7	7.9	9.4	12.0	14.9	15.8	14.7	14.1	11.3	7.9	4.7	(22)	(23)
	5%	DB	12.5	13.9	19.1	21.2	25.0	32.2	32.6	32.3	30.1	22.9	17.4	12.4	(23)	(24)
		MCWB	3.7	4.4	7.0	8.8	11.8	14.6	15.7	14.9	13.5	9.8	6.5	3.6	(24)	(25)
	10%	DB	10.9	11.3	17.0	18.7	22.4	29.9	31.2	31.0	27.7	21.1	15.1	10.1	(25)	(26)
		MCWB	3.0	3.2	5.8	8.0	11.3	14.2	15.7	14.9	13.1	9.5	5.3	2.6	(26)	(27)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	7.9	9.8	11.2	15.9	19.4	18.9	18.5	16.8	13.4	9.2	5.9	(27)	(28)
		MCDB	16.8	19.3	24.2	21.1	23.0	27.5	25.8	24.6	24.2	23.8	21.3	16.1	(28)	(29)
	2%	WB	5.3	5.9	8.4	10.3	14.3	17.8	18.1	17.6	15.8	12.3	8.0	4.8	(29)	(30)
		MCDB	14.2	15.8	20.3	20.6	23.2	26.7	25.8	25.8	23.8	21.8	18.6	14.0	(30)	(31)
	5%	WB	4.0	4.8	7.2	9.5	13.0	16.6	17.7	16.8	15.1	11.3	6.9	3.8	(31)	(32)
		MCDB	11.9	13.6	17.9	19.4	20.8	25.5	25.7	25.7	24.1	20.8	16.4	12.0	(32)	(33)
	10%	WB	2.8	3.5	6.1	8.6	12.1	15.7	17.0	16.2	14.4	10.1	5.8	2.6	(33)	(34)
		MCDB	10.2	10.8	15.4	17.6	20.0	24.9	25.4	25.5	24.3	19.4	14.2	9.7	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	11.3	12.7	12.7	12.1	14.5	14.3	14.6	14.3	13.3	12.4	11.3	(35)	(36)
		MCDBR	15.2	14.7	16.4	16.8	16.5	17.7	16.7	17.0	17.9	18.0	16.5	14.1	(36)	(37)
	5% WB	MCWBR	8.0	7.3	7.4	7.1	6.6	6.1	5.1	4.8	5.8	7.2	7.7	7.5	(37)	(38)
		MCWBR	14.7	14.2	15.7	15.0	13.9	14.7	13.7	14.0	14.2	15.2	15.6	13.8	(38)	(39)
(40) Clear-Sky Solar Irradiance	taub		0.244	0.253	0.278	0.306	0.326	0.333	0.365	0.356	0.313	0.279	0.258	0.246	(40)	(41)
	taud		2.577	2.545	2.499	2.464	2.474	2.468	2.423	2.435	2.541	2.612	2.590	2.577	(41)	(42)
	Ebn at Noon		954	986	986	970	950	939	905	907	933	942	926	923	(42)	(43)
	Edh at Noon		68	82	96	107	109	110	114	109	92	76	67	63	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		2.59	3.60	4.80	5.55	6.06	7.02	6.50	5.78	5.11	3.76	2.69	2.21	(44)	(45)
	RadStd		0.12	0.20	0.39	0.38	0.49	0.37	0.30	0.25	0.30	0.24	0.16	0.14	(45)	(46)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page